SAMUEL J. TUSICK

724-601-4365 | samtusick@outlook.com
linkedin.com/in/samuel-tusick | samtusick.github.io/Portfolio-Website

PROFESSIONAL SUMMARY

Software engineering student (May 2027) working in an agile environment building AI and automation tools used daily by an institute, co-designing an LLM-driven virtual patient platform tested by health professionals, and leading a 4-person student team building a machine learning payload for a rocket flight. Led four student project teams. U.S. citizen, eligible to pursue a security clearance.

EDUCATION

Florida Gulf Coast University, B.S. Software Engineering *Fort Myers, FL | Expected May 2027*

- In-Major GPA: 3.83 | Dean's List: Fall 2023, Spring 2024, Fall 2024, Spring 2025
- Honors: Ruth M. and Cleveland L. Campbell Engineering Scholarship Endowed Fund, Kunkle Family Scholarship Endowed Fund
- Relevant Coursework: Machine Learning, AI Agent Studio, Software Architecture & Design, Cloud Computing, Intro to Data Engineering, Data Structures & Algorithms, Software Engineering Fundamentals, Requirements Engineering & Analysis

TECHNICAL SKILLS

- **Languages:** Python, C++, SQL, Java, JavaScript, TypeScript
- **Applied AI and Automation:** LLM integration, prompt engineering, agentic workflow automation, retrieval-augmented generation (RAG), multi-service AI orchestration, neural networks and genetic algorithms
- **Cloud and Backend:** AWS (API Gateway, Lambda, Aurora, Cognito, IAM, CDK, Transcribe), Docker, Flask, REST APIs, PostgreSQL (pgvector), Supabase, SQLite
- **Frontend:** React, Vite, Tailwind CSS
- **Tools:** Git, GitHub Actions, Selenium, TestNG, Jira, Confluence, Azure DevOps, Power Automate, NumPy, Pygame, Gymnasium, OpenRocket, Agile

EXPERIENCE

FGCU Dendritic AI & Data Science Institute, Student Research Assistant *Mar 2025 to Present | On-site*

- Cut course module production time about 50% (about 90 to 45 minutes) by building two LLM tools: one turns faculty PDFs and slides into a reviewed narration script, and the other turns the approved script into an avatar video and slide deck. A cost estimate and approval step runs before any credits are spent, so one script review replaces manual scripting, video setup, and slide building. Used by the development and infrastructure teams.
- Created the institute's first workflow handbook (25 pages, used by all 10 team members) with a tool that writes and updates Confluence pages in one fixed template, checks for duplicates, screens out private data, then validates each page and reads it back to confirm it published correctly. Every page puts the owner, steps, tools, and review rules in the same place.
- Gave the whole team one standup-style view every morning by automating a daily Jira summary posted as a Microsoft Teams card at 9 AM, covering every task on the board, running since October 2026.
- Replaced texting between teams, where updates were often lost, with Power Automate alerts that post to Teams when files land in about 25 monitored folders, for the Digital Media and Design and Infrastructure teams.
- Built an AI tool that turns a topic into a finished 1080p explainer video with voiceover in about 25 minutes, replacing a day or two of Digital Media and Design team time, using the team's exact content and branding guidelines. It drafts a script for review, generates a voiceover with word-level timestamps, builds an animation timed to them, and renders in parallel with the audio mixed in.
- Helped plan and design the 2026 FGCU Summer AI Academy (500+ participants, 150+ instructional hours), ran day-to-day operations, and taught a GitHub Actions lesson in the Software Testing course.
- Built 14 custom AI tutors covering 13 engineering courses for the Engineering Learning Hub, used by students seeking tutoring help and by Learning Hub tutors.

Router-Level Security Detection, Subteam Lead and Backend Developer *Feb 2025 to Apr 2025 | Lab class group project*

- Led backend architecture decisions and task coordination for a 5-member team, delivering the project by its April 29 deadline.
- Built a Flask web application that gathers CPU, memory, bandwidth, and connected-device data from 3 router models over Secure Shell using 7 analysis commands, and serves it through a REST application programming interface (API) to a dashboard. Stored results in SQLite, and tested and deployed the router-communication scripts.

PROJECTS

MAIVA: AI Virtual Patient Training Platform | TypeScript, JavaScript, React, AWS, LLM API

- Co-designed, with a graduate researcher, a platform where students practice clinical interviews with AI-simulated patients, built from requirements set by two clinical assistant professors. Chat and voice modes are working and have been tested by 3 professionals, including Marieb's dean. Instructors build their own patient scenarios and every transcript is saved for review.
- Co-designed the orchestrator that connects the interface, speech-to-text, text-to-speech, the LLM API, and AWS services, streaming each patient reply sentence by sentence, saving every turn in real time, and rebuilding a session from its saved transcript after a restart.
- Built the React frontend from scratch, integrated AWS Transcribe for speech input, and wrote the infrastructure endpoints and CDK in TypeScript plus the SQL data layer.
- Co-built Local, Development, and Test environments on a serverless AWS backend (API Gateway, Lambda, Aurora, Cognito, IAM,

CDK).
- Avatar mode is in development: 3D patient models are built in Unreal Engine 5, and the team is designing how to drive their facial animation from text-to-speech viseme data.

Real-Time Landing Site Detection with Machine Learning | Python, OpenRocket
- Team lead for a 4-person team selected as one of two winning teams in the FGCU Payload Challenge, earning $500 in funding and a flight on the club rocket, earliest January 2027.
- Ran a proof of concept with a pretrained segmentation model that proposed 3 landing zones with a safety percentage for every frame; now planning to train our own model on aerial datasets and run it on a Raspberry Pi with a 4K camera.
- Chose to test during powered ascent rather than parachute descent, to stress the model with flame, vibration, spin, and motion blur. Set targets of at least 2 predictions per second and 80% pixel accuracy, to be reported whether met or not.
- Built an OpenRocket flight simulation of the club rocket.

Email Summary Agent (personal project) | Python, AWS Lambda, Docker, PostgreSQL (pgvector), Microsoft Graph API
- Built and deployed an agent that runs daily across 2 Outlook accounts, summarizing every email from the last 24 hours with an urgency level using retrieval over embedded email content. Runs on a schedule in a Docker container on AWS Lambda, with failure alerts and secrets in AWS Secrets Manager, and data in Supabase.

ML Self-Driving Car Simulator | Python, Pygame, Gymnasium
- Building a neural network and genetic algorithm trainer from scratch in Python, with no machine learning libraries, to teach a car agent to complete a track; now tuning mutation rate, sigma, and pool fraction.
- Built the simulation: steering-based car physics, pixel-mask collision detection, five ray-cast distance sensors, a checkpoint system, and a Gymnasium environment.

Switch4Good Database System | PostgreSQL, SQL
- Led a 3-person class team that designed a 19-table PostgreSQL database with 3 access roles for the nonprofit Switch4Good, sized for its 100+ universities and 1,500+ students and now used by the client.
- Led client communication, assigned tasks, ran reviews, and wrote most of the documentation, schema, and design work.

Pinterest Testing Suites | Java, Selenium, TestNG
- Led a 2-person team that wrote 46 test cases across 8 modules (functionality, usability, regression), personally authoring 25, with a 97.8% pass rate (45 of 46).
- Resolved Pinterest's limited login attempts by restructuring the login test timing.

NHL Stat Scraper | Python, Flask, Selenium, React, Vite
- Built a Selenium-based Flask scraper covering 15+ stat categories for every NHL player, serving fresh results on every request through REST APIs to a React and Vite frontend with filterable views. Open source on GitHub.

LEADERSHIP AND INVOLVEMENT

Deeptech Club, Payload Team Lead *Oct 2026 to Present*

Deeptech Club, Member *June 2026 to Present*

Computer Science & Software Engineering Club, Member *Aug 2024 to Present*

FGCU AI Day, Panelist, "AI in University Operations" *Mar 2026*

Led four student project teams of 2 to 5 members: Eagle CyberNest, Switch4Good, Pinterest Testing, Landing Site Payload*2025 to Present*

Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate (valid until Aug 29, 2027) *Aug 2025*